Serum PBDEs and Age at Menarche in Adolescent Girls: Analysis of the National Health and Nutrition Examination Survey 2003–2004

BACKGROUND Polybrominated Diphenyl Ethers (PBDEs), widely used as flame retardants since the 1970s, have exhibited endocrine disruption in experimental studies. Tetra- to hexa-BDE congeners are estrogenic, while hepta-BDE and 6-OH-BDE-47 are antiestrogenic. Most PBDEs also have antiandrogenic activity. It is not clear, however, whether PBDEs affect human reproduction. OBJECTIVES The analysis was designed to investigate the potential endocrine disruption of PBDEs on the age at menarche in adolescent girls. METHODS We analyzed the data from a sample of 271 adolescent girls (age 12–19 years) in the National Health and Nutrition Examination Survey (NHANES), 2003–2004. We estimated the associations between individual and total serum BDEs (BDE-28, -47, -99, -100, -153, and -154, lipid adjusted) and mean age at menarche. We also calculated the risk ratios (RRs) and 95% confidence intervals (CI) for menarche prior to age 12 years in relation to PBDE exposure. RESULTS The median total serum BDE concentration was 44.7 ng/g lipid. Higher serum PBDE concentrations were associated with slightly earlier ages at menarche. Each natural log unit of total BDEs was related to a change of –0.10 (95% CI: –0.33, 0.13) years of age at menarche and a RR of 1.60 (95% CI: 1.12, 2.28) for experiencing menarche before 12 years of age, after adjustment for potential confounders. CONCLUSION These data suggest high concentrations of serum PBDEs during adolescence are associated with a younger age of menarche.